

Review

Influence of rice husk ash (RHA) on the properties of self-compacting concrete: A review



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HIGHLIGHTS

- Paper reviews the fresh and hardened properties of RHA-SCC.
- Incorporation of RHA improved the workability and self-compactibility.
- Cement replacement by 10–15% RHA improved the strength and permeation properties.
- Addition of mineral admixtures in RHA-SCC improved resistance to acid and ASR.

ARTICLE INFO

Article history:

Received 20 July 2016

Received in revised form 2 June 2017

Accepted 22 July 2017

Available online 28 July 2017

Keywords:

Durability

RHA

Self-compacting concrete

Strength

SEM

XRD

ABSTRACT

Solid waste management of waste materials or byproducts is a serious environmental issue. One such byproduct is rice husk ash (RHA). It is obtained by burning of rice husk and produces reactive amorphous silica which contains approximately 90% silica. The pozzolanic nature of RHA due to high silica content makes it a valuable supplementary cementitious material (SCM) for utilization in cement-based materials. RHA can be used in making self-compacting concrete (SCC). Research in this direction has been reported by several researchers. In this paper, review of the work done on the RHA's physical, chemical properties, SEM and XRD analysis, and effect of RHA on fresh, strength and durability of SCC is presented. Due to its pozzolanic nature, RHA significantly influences the properties of Self-Compacting concrete. Incorporation of 10–15% RHA as partial replacement of cement enhances strength and durability properties of SCC. Research on the role of RHA in SCC, will not only make its utilization in SCC, but will reduce land-filling costs and also provides a cleaner sustainable environmental solution in saving energy and reducing carbon dioxide generation by cement consumption.

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1. Introduction

World-wide yearly production of rice is 742 million tons, and approximately 148 million tons of rice husk is produced [1]. For every ton of husk, approximately 0.19 ton of ash is generated. Rice husk ash (RHA) is produced by burning rice husk. Generally, every ton of husk produces about 0.19 ton of ash [2,3]. Rice husk has a high calorific value [4,5]. Amorphous silica is mostly concentrated at the surfaces of the rice husk and not within the husk itself [6]. RHA is an agricultural by-product available in large quantities causing detrimental effect due to crop residue burning. By utilization of RHA in SCC greener and economical concrete is achieved. RHA an agriculture by-product being in abundance, having high surface area and pozzolanic in nature, can be effectively used as supplementary cementitious material (SCM). SCC is High Performance Concrete which saves time and energy during construction. Using RHA as SCM as partial replacement of cement helps to achieve greener and economical concrete as it reduces the cost of SCC. Further, use of RHA results in environmental benefit as lesser quantity of it would have to be land-filled, leading to reduction in environmental pollution.

Carbonization and decarbonation are two different phases in the decomposition of rice husk. Silica in RHA melts around 1440 °C [5]. Burning of rice husk at temperature lower than 800 °C produces reactive amorphous silica which contains approximately 90% silica, whereas above this temperature some crystalline form of silica can also be obtained [7–10]. Silica in amorphous form is useful as a pozzolan to produce durable good quality concrete [8]. Crystalline and amorphous forms of silica in rice husk depend upon burning temperature and its duration. Rice husk ash should be produced with suitable specifications for a specific use as crystalline and amorphous forms of silica have different properties [4,11,12]. Nair et al. [13] stated that the samples burnt for more than 12 h, at 500–700 °C produced high reactivity ashes, whereas short durations burning (15–360 min) resulted in high carbon content in RHA. Mehta [14] reviewed physical and chemical properties of RHA, and its use as a supplementary cementing material. Use of RHA in cement based materials reduces heat of hydration, improves strength and durability parameters

besides reduction in cement cost and other environmental benefits.

1.1. Research gaps

RHA can be effectively utilized up to 10–15% as partial replacement of cement. There is lot of published literature on use of RHA on strength properties of concrete/SCC, however there is limited research available on durability aspect such as shrinkage, creep and abrasion and Alkali silica reaction, etc. Therefore research need to be directed towards these properties. Further, higher strength concrete using RHA and nano particles can be investigated so that SCC with high strength can be designed.

1.2. Properties of RHA

1.2.1. Physical properties

RHA is a fine material having grey to white in color due to complete burning while partially burnt RHA is blackish. Table 1 shows physical properties of RHA.

Della et al. [15] reported after burning RHA samples at 700 °C for 6 h and wet-grinding (80 min), reduced particle size and grey colour was obtained because of lower carbon content (Fig. 1a). The carbon content in RHA before burning was 18.60% which was reduced to 0.14% (Fig. 1b). Average particle size of rice-husk ash varied between 3 and 10 µm. Before burning, RHA had a specific surface area of 177 m²/g, which got reduced 54 m²/g after burning, and then increased to 81 m²/g after wet grinding.

Habeeb and Mahmud [16] observed that with increasing grinding time from 90 to 360 min, size of RHA decreased from 63.8 to 11.5 µm (Fig. 2) with slight increase in specific surface area with time due to micro-porous and multilayered surface of RHA. Della et al. [15] reported that after burning RHA at 700 °C for 6 h, its mean particle size was 33 µm (Fig. 3), which after milling for 80 min, reduced to 0.68 µm (Fig. 3).

1.2.2. Chemical composition and SEM/XRD of RHA

The chemical composition of rice husk varies due to variations in quality of rice, and generally contains cellulose (40–50%), lignin

Table 1
Physical properties of RHA.

Property	Kannan and Ganesan [38]	Le et al. [63]	Sua-iam et al. [32]	Safiuddin et al. [29]	Habeeb and Mahmud [16]	Ganesan et al. [64]
Mean particle size (µm)	6.27	5.7–15.6	39.34	6	11.5–63.8	3.80
Specific gravity	2.08	–	2.24	2.1	–	2.06
Fineness: passing 45 µm (%)	91	–	–	–	–	99
Specific surface area (m ² /g)	36.47	22.36–25.21	0.37	2.33	25.3–30.4	36.47

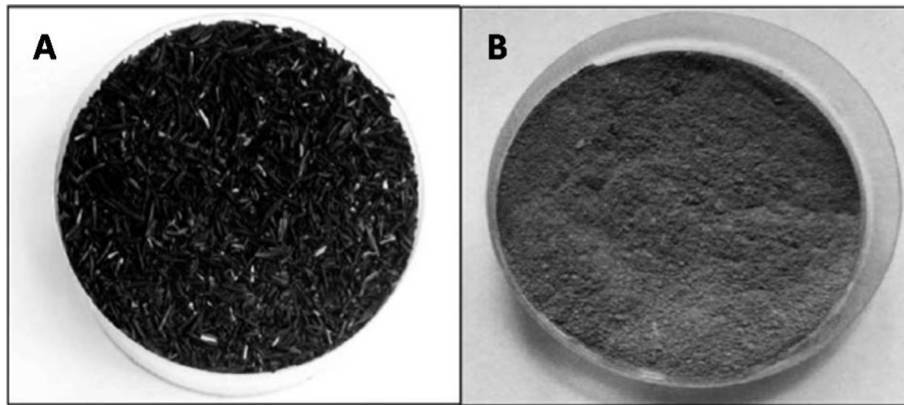


Fig. 1. Rice husk ash, (A) as received, (B) after burning out at 700 °C for 6 h [15].

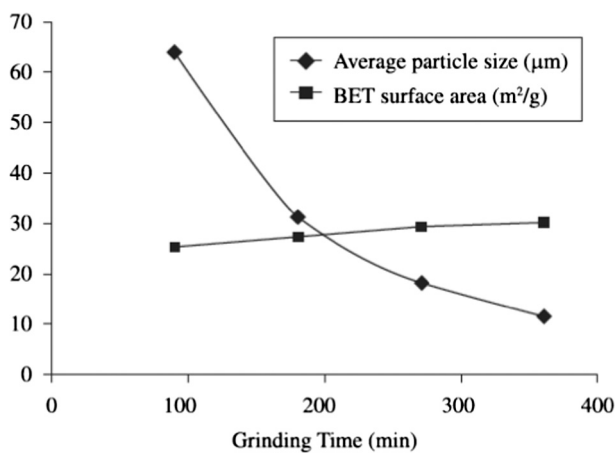


Fig. 2. Particle size and specific surface area of RHA versus grinding time [16].

(25–30%), ash (15–20%) and moisture (8–15%) [17]. Chemical properties of rice husk are tabulated in Table 2. Silica Oxide is the main component in RHA, and it varies between 77.19 and 94.95%. Stroeve et al. [12] found that the silica is distributed mostly under the husk's outer surface. Silica content in RHA ranges from 85 to 95%.

X-ray diffraction (XRD) analysis of RHA exhibited that it consists of amorphous materials [3] and shows broad peak on 2θ angle of 22° (Fig. 4a) [16]. Farooque et al. [18] also analyzed the XRD of RHA and indicated the presence of quartz (22.85° , 26.63° , 42.47° 2θ), cristobalite (21.91° , 35.99° , 69.5° 2θ) and anorthite (27.91° , 29.42° 2θ) peaks (Fig. 4b) [18]. Presence of cristobalite peak indicated the crystalline nature of silica in RHA produced at high temperature (above 973 K). Cizer et al. [19] reported amorphous phase

of RHA with a broad band between 15 and 30° 2θ (Fig. 4c), along with specific quantity of crystalline silica in cristobalite and tridymite form. This established that RHA was obtained by burning at 800–1000 °C for crystallization of the amorphous silica takes place.

SEM of rice husk ash shows silicious nature and highly porous tracery surface morphology, with a high surface area. Farooque et al. [18], Habeeb and Mahmud [16] and Madrid et al. [20] analyzed the RHA particles by SEM and showed multilateral, angular, microporous surface and honeycombed structure of RHA justifying its high specific surface area (Fig. 5). Habeeb and Fayyadh [21] and Chindaprasirt et al. [8] revealed that the rice husk after burning at 700 °C maintains its cellular structure and after grinding, particles shape are very irregular- with porous cellular surface. Rego et al. [22] reported that RHA with higher amorphous silica formed the interstitial pores smaller than 500 nm, due to aggregation of the non-condensed SiO_2 particles inside the voids in RHA. Xu et al. [23] found that interstitial pores in RHA were dependent on the calcination process and greatly influenced its specific surface area. After 5 h of grinding of RHA, Fig. 6 represents its surface and morphology at different magnification levels showing macro and interstitial pores filled with finely grounded particles. At high magnification level of (20,000 $\times$) interstitial pores within of macropores can be seen. Ranjbar et al. [24] carried out XRD analysis and showed that intensity of Alite and Belite phases decreased and new peak of portlandite achieved with the addition of nano- TiO_2 (NT). The SEM micrographs illustrated the widespread distribution of mortars containing NT with packed pore structures which resulted in promoting of strength and durability of specimens.

X-ray diffraction analysis indicates that the RHA contains mainly amorphous materials with a small quantity of crystalline phases such as cristobalite (high-temperature phase of SiO_2) and sylvite (KCl), which is probably originated from the use of the fertilizers.

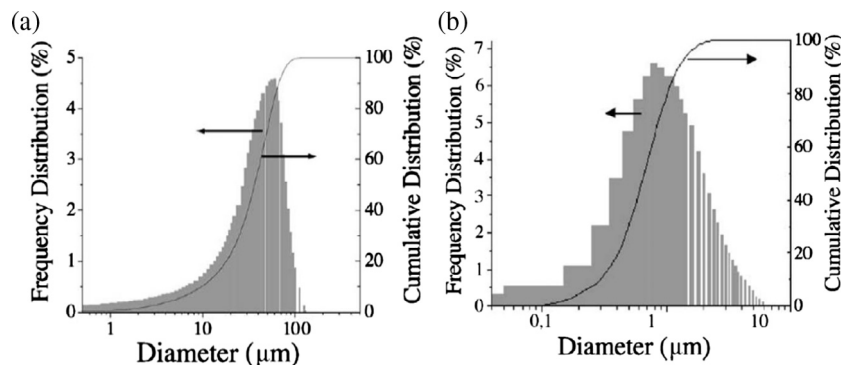


Fig. 3. Particle size distribution of RHA [15].

Table 2
Chemical properties of RHA.

Constituents	Rego et al. ^a [22]	Kannan and Ganesan [38]	Chopra et al. [40]	Sua-iam and Makul [32]	Memon et al. [31]	Della et al. ^b [15]
Silica (SiO ₂)	87.08	87.89	94.0	93.44	77.19	94.95
Aluminium oxide (Al ₂ O ₃)	<0.01	0.19	1.2	0.21	6.19	0.39
Iron oxide (Fe ₂ O ₃)	0.11	0.28	0.37	0.18	3.65	0.26
Calcium oxide (CaO)	0.70	0.73	2.93	0.76	2.88	0.54
Magnesium oxide (MgO)	0.42	0.47	0.60	0.43	1.45	0.90
Sodium oxide (Na ₂ O)	0.18	0.66	–	0.05	–	0.25
Potassium oxide (K ₂ O)	1.40	3.43	0.50	1.98	1.82	0.94
Sulfur oxide (SO ₃)	–	–	0.30	0.16	–	–
LOI	8.03	4.36	–	1.27	5.429	0.85

^a RHA sample grinded for 5 h.
^b RHA after burning out.

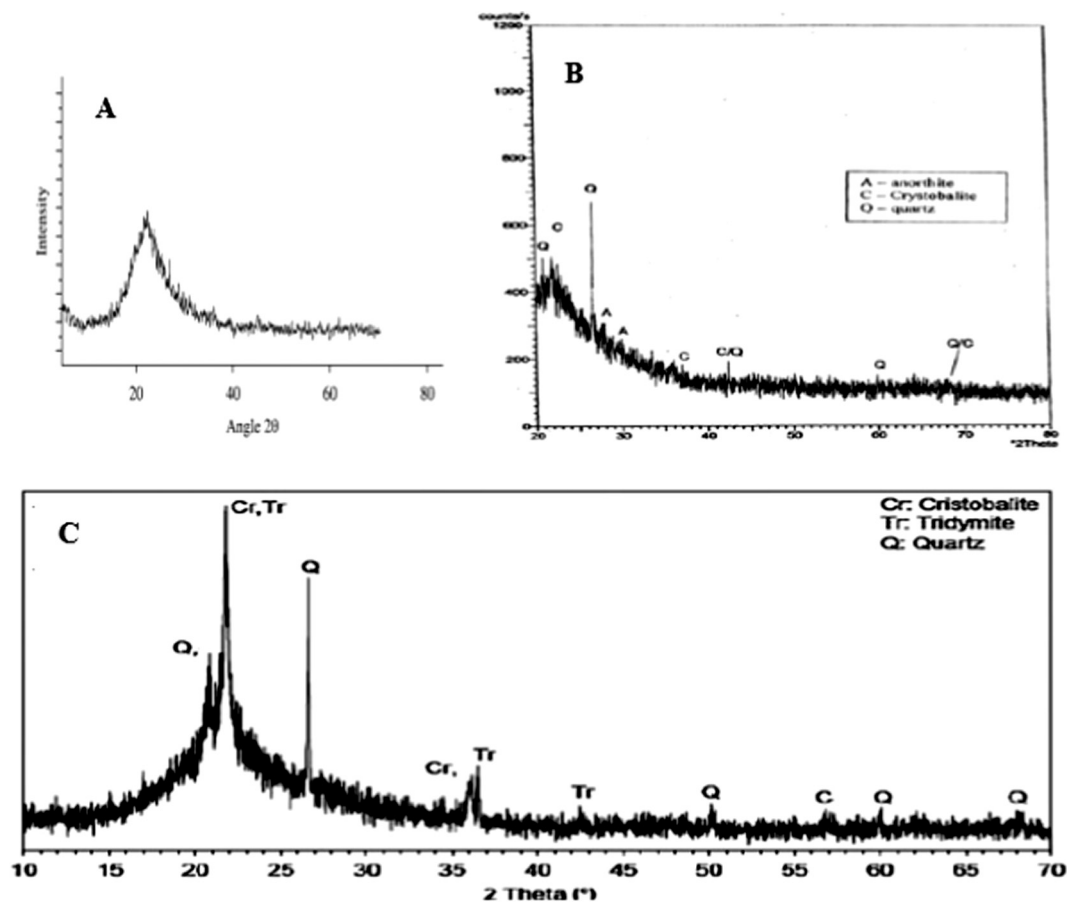


Fig. 4. XRD of Rice husk ash, (A) [16], (B) [18], (C) [19].

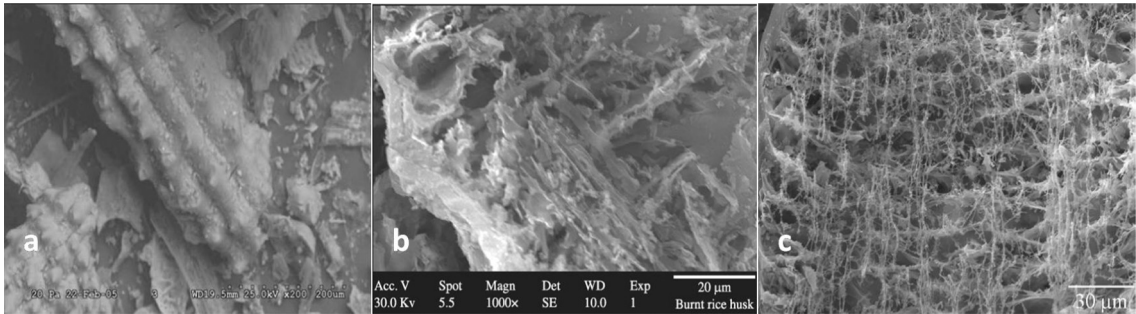


Fig. 5. SEM of RHA, (a) [18], (b) [16], (c) [20].

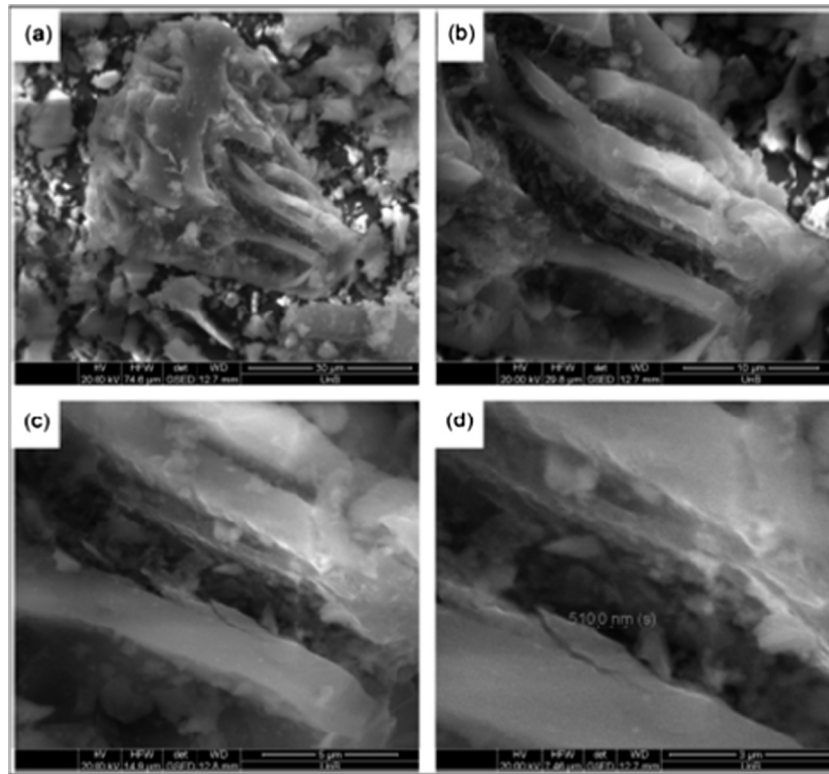


Fig. 6. Surface structure and morphology of the RHA after 5 h of grinding: (a) RHA particle at low magnification (2000 $\times$), (b) RHA particle at (5000 $\times$), (c) RHA particle with magnification (10,000 $\times$), (d) RHA particle with magnification (20,000 $\times$) [22].

1.3. Applications of RHA

Rice husk ash can be used in following applications (Krishna, [25]; Siddique & Khan, [26]; Ricehuskash.com, [27]; Pode et al. [28])

- Blended cements
- Green concrete
- High performance concrete
- Refractory
- Ceramic glaze
- Insulator
- Roofing shingles
- Waterproofing chemicals
- Oil spill absorbents
- Carrier for pesticides, biofertilizers
- solar panels
- Plastic and rubber reinforcements
- Catalysts, Coatings
- Pulp and paper processing
- Detergents and soap
- Anticaking agent for packing

1.4. Advantages of using RHA in cement and concrete

Rice-husk ash, a very fine pozzolanic material, when blended with cement makes it the most versatile eco-friendly supplementary cementitious material to concrete. The utilization of rice husk ash as a pozzolanic material in cement and concrete provides several advantages such as:

- Reduced heat of hydration
- Improved strength
- Reduced permeability at higher dosages

- Increased chloride and sulphate resistance/mild acids
- Reduced materials costs due to cement savings, and
- Environmental benefits related to the disposal of waste materials and to reduced carbon dioxide emissions.

2. Effect of RHA on fresh properties of SCC

2.1. Slump

Safiuddin et al. [29] studied the slump of SCC mixtures made with 5–30% RHA; 3.5–4.5% super plasticizer and varied w/c ratio of 0.30–0.40. All the mixes exhibited slump in the range of 265–280 mm. Though the slump increased by only 5 mm in 30% RHA compared to 0% RHA (270 mm) at 0.35 w/c ratio, but the deformability of concrete was significantly improved. Ludwig et al. [30] studied physical properties of RHA and found that the pore size distribution is the key parameter controlling the pore volume, Specific surface area (SSA) and water demand of RHA which strongly influence the rheological behaviour and flowability of mortar. Increasing content of RHA increases the SP saturation dosage of mortar. The lower the w/b ratio is, the higher the saturation SP dosage of mortar. The incorporation of RHA decreases the mini slump flow, and increases the yield stress and plastic viscosity. This effect is much stronger when the coarser RHA or the higher content of RHA is used.

2.2. Slump flow

Memon et al. [31] studied the fresh properties of SCC incorporating 5% and 10% RHA; and superplasticizer content (3.5–4.5%), and concluded that slump flow for all the mixes except mix with 10% RHA and 3.5% super plasticizer (595 mm) were within the EFNARC range (650–800 mm) of SCC. Increase in superplasticizer content increased the flow whereas flow decreased with increase

in RHA quantity. Safiuddin et al. [29] observed slump flow of SCC mixtures in the range 665 mm to 770 mm and found relative difference in different mixtures. SCC mixtures having 0 and 30%RHA achieved slump flow between 690 and 750 mm at w/p ratio of 0.35.

Sua-iam and Makul [32] prepared several SCC mixtures containing RHA (0–100%) as fine aggregate replacement, and reported that with increase in RHA content, slump flow was also increased, exhibiting slump flow of 70 ± 2.5 cm. Higher slump flow at higher RHA percentages (80 and 100%) was due to enhanced surface area of RHA, which resulted in increased viscosity of RHA-SCC. In another study, conducted by Sua-iam and Makul [33], slump flow time increased with increase in RHA content. Mixtures containing 0–60% coal fly ash and up to 75% RHA content showed slump flow time within 3–7 s and in the acceptable range of EFNARC [34]. Mixtures containing 100% RHA showed higher slump flow time as they absorbed free water because of its higher porosity and angular shape. Addition of RHA (331.3 kg/m^3) as supplementary cementitious material in SCC showed slump flow of 710 mm, and falls within the acceptable range of EFNARC [35].

Atan and Awang [36] found that replacement of 15% ordinary Portland cement (OPC) by raw RHA (RRHA; specific gravity 2.16 and, Blaine fineness $351 \text{ m}^2/\text{kg}$) increased the water requirement of mix by 38%. RHA absorbs large quantity of water on the surface because of its higher specific resulting in less water available for lubrication and thus reducing the flowability. Ternary mixes of SCC containing OPC, RHA (15%) and equal mass of limestone (LP), pulverized fuel ash (FA) and silica fume (SF) showed similar requirements of water at 30% replacement level as in binary mixes. In quaternary mixes (OPC + RHA + FA + LP and OPC + RHA + SF + LP), at 45% OPC replacement level, reduction in super plasticizer requirement was observed whereas mix containing OPC + RHA + SF + FA showed higher requirement for water and super plasticizer. This was due to the extreme fineness of SF particles (Blaine fineness $20,000 \text{ m}^2/\text{kg}$) that coupled with RHA particles resulted in high surface which increased the viscosity of the mix [36,37].

Kannan and Ganesan [38] studied the slump flow of SCC with RHA (0–30% with 5% increment) and compared with EFNARC [39] recommended guidelines (Table 3). All RHA-SCC mixes (550–740 mm) except 30% RHA-SCC mix (495 mm) are in the range of slump flow classes 1 (SF1) and 2 (SF2). It was concluded that with increase in RHA content, slump flow decreased primarily due to higher reactivity and high surface area of RHA. Similar findings were also observed by Chopra et al. [40] where the slump flow ranged between 600 and 730 mm for all the RHA-SCC mixtures. Le and Ludwig [41] investigated that use of RHA increased the SSD (superplasticizer saturated dosage) of mortar, slightly decreased filling and passing abilities and significantly increased plastic viscosity and segregation resistance of self-compacting high performance concrete (SCHPC). The bleeding of mortar and SCHPC was not seen in concrete incorporating RHA. At higher replacement levels of RHA the effect was more prominent. RHA with macro-mesoporous structure improves viscosity of SCC varying dosage of superplasticizer. RHA particles have large specific surface area, coarse particles create more affinity for water and superplasticizer.

2.3. L-Box test

It measures the passing and filling ability of SCC. Memon et al. [31] found that SCC mixes (5 and 10% RHA; 3.5–4.5% super plasticizer) achieved L-box ratio of 0.8–10.0, which is within the ranged as prescribed by EFNARC [34]. L-Box ratio decreased with increase RHA content whereas it increased with increase in superplasticizer content. Pai et al. [35] achieved 0.86 of L-box value for SCC mix containing RHA (331.3 kg/m^3) in addition to cement (200 kg/m^3). Kannan and Ganesan [42,38] observed the L-box blocking ratio of different RHA-SCC mixtures according to EFNARC [39] specifications (Table 3). The L-box values for all the mixtures varied from 0.59 to 0.94, and for 15% RHA-SCC mix satisfactory blocking ratio was observed, while the L-box values were ranged between 0.8 and 1 in SCC mixes with up to 15% RHA content in the study conducted by Chopra et al. [40]. The blocking ratio for other mixes did not satisfy EFNARC recommended values (Table 4).

2.4. V-Funnel test

Flowability and segregation resistance of SCC mixtures is evaluated by V-funnel test. Higher flowability of mixtures indicates shorter flow time. According to EFNARC [34], the V-funnel test ranges from 6 to 12 s. Memon et al. [31] observed that most of the V funnel test values were less than 6 s, which exhibited the filling ability of the SCC mixes. Increase in quantity of RHA (10%) enhanced the viscosity of mix. At lower superplasticizer content (3.5%), 5% RHA content showed V-funnel test within the EFNARC range, while at 4 and 4.5% super plasticizer content, 10% RHA qualified the V-funnel test.

Similar findings were also observed by Sua-iam and Makul [32] in SCC mixtures containing RHA as replacement to fine aggregate. Mixes containing 10 and 20% RHA showed acceptable flow time of 9 and 11 s, respectively, whereas increasing RHA content (40, 60, 80 and 100%) increased the flow time as RHA particles absorbed the water, resulted in viscous mix and reduced bleeding. Study conducted by Sua-iam and Makul [27] reported that V-funnel flow time was within 8–12 s when SCC was made with RHA content less than 25% (Table 4). With increasing RHA content the V-funnel flow time increased due to increased viscosity, and this was improved by incorporation of fly ash particles. Pai et al. [35] reported acceptable range of V-funnel flow value (9 s) for SCC containing RHA (331.3 kg/m^3) in addition to cement (200 kg/m^3). Kannan and Ganesan [38] observed the V-funnel time values for different RHA-SCC mixes varied in the range 3.9–8.4 s, and categorized under VF1 class according to EFNARC [39] guidelines (Table 3).

2.5. Orimet test (flow time and flow spread)

As per EFNARC [34], higher limit for orimet flow time is 9 s Safiuddin et al. [29] reported the orimet flow time of RHA (10–30%) incorporated SCC from 4.8 s to 11.5 s. Higher the flow values of orimet time indicates higher viscosity of concrete. At low w/c ratio (0.30) and with increasing RHA content (25 and 30%) orimet flow time was slightly above the EFNARC limit. This suggested the viscous nature of these mixes.

Table 3
Recommended values for fresh state properties of SCC according to EFNARC [39]

Slump flow test		V-funnel test		L-box test	
Slump flow classes	Slump flow (mm)	Viscosity classes	V-funnel times (s)	Passing ability classes	Blocking ratio (H2/H1)
SF1	550–650	VF1	≤8	PA1	≥0.8 with 2 bars
SF2	660–750	VF2	9–25	PA2	≥0.8 with 3 bars
SF3	760–850	–	–	–	–

Table 4

Fresh properties of RHA-SCC mixes studied by several researchers.

Study	RHA replacement (%)	w/b	Super plasticizer content (%)	Fresh concrete properties			
				Slump flow (mm)	L-box (H2/H1)	V-funnel (s)	J-ring (mm)
EFNARC [34]	–	–	–	650–800	0.8–1	6–12	–
Chopra et al. [40]	0	0.41	1	730	1	6	–
	10			700	0.9	8	–
	15			670	0.8	11	–
	20			600	Blocking	13	–
Kannan and Ganesan [42]	0	0.55	2	740	0.94	3.9	–
	5			700	0.93	4.0	–
	10			670	0.87	4.0	–
	15			610	0.82	4.8	–
	20			580	0.78	6.0	–
	25			550	0.71	7.2	–
	30			495	0.62	8.0	–
Rahman et al. [54]	0	0.38	1.8	630	–	5.9	5.2
	20	0.5	3.5	660	–	6.6	3.7
	30			670	–	6.3	3.5
	40			580	–	7.0	4.4
Sua-iam and Makul [33]	0	0.25	1.2 (HRWRA)	720	–	10	720
	25	0.54		700	–	12	690
	50	0.90		700	–	34	660
	75	1.46		680	–	72	630
	100	1.89		680	–	–	580
Safiuddin et al. [29]	0	0.30	0.87	710	–	–	730
	15		1.75	735	–	–	745
	20		2.10	770	–	–	765
	0		0.35	690	–	–	690
	5	0.35	0.87	700	–	–	695
	10		1.05	710	–	–	705
	15		1.40	720	–	–	720
	20		1.75	710	–	–	715
	25	0.40	2.10	740	–	–	760
	30		2.45	750	–	–	770
	0		0.60	665	–	–	675
	15		1.00	680	–	–	700
	20		1.20	675	–	–	705
Memon et al. [45]	0	0.4	3.5	770	1	6	–
	5	0.38		650	1	6.8	–
	10	0.36		595	Blocking	29.3	–
	0	0.4		780	1	4.24	–
	5	0.38	4.0	710	1	4.26	–
	10	0.36		660	0.88	11.8	–
	0	0.4		795	1	2.18	–
	5	0.38		760	1	3.93	–
	10	0.36	4.5	700	0.94	8.62	–

2.6. Passing ability of concrete

J-ring slump, slump cone- J ring flow spread, orimet- J ring flow tests were conducted to evaluate the passing ability of SCC mixtures. Safiuddin et al. [29] found that the addition of adequate dose of high range water reducer (HRWR) helps to achieve passing ability of SCC. The J-ring slump values were between 255 and 270 mm for various SCC mixtures, whereas slump cone J-ring flow spread was between 650 and 740 mm which were lower than the slump flow by 15–30 mm. Similarly, the orimet J-ring flow spread ranges in 675–770 mm and was decreased by 15–30 mm when compared to orimet flow spread. Incorporation of RHA in SCC as fine aggregate replacement exhibited small degree of blocking which can be improved by using finer particles such as lime stone [32] and fly ash [33]. Table 4 shows the fresh properties of SCC mixes containing RHA studied by several researchers.

Published literature indicates that to satisfy the requirements of 'filling and passing ability and resistance to segregation' optimum RHA content was found to be between 10 and 20%. It is dependent on w/b ratio and dosage of superplasticizer. High RHA volume tend to decrease both workability and segregation. RHA could be used at

a level of up to 25% in SCC and fulfill the criteria according to EFNARC.

3. Strength properties of SCC

3.1. Compressive strength

Ahmadi et al. [43] studied the compressive strength of SSC mix made with RHA (10 and 20%) in comparison to normal concrete at two w/b ratios (0.40 and 0.35) up to the age of 180 days. SCC mixes achieved higher compressive strength (31–41%) than normal concrete. Sua-iam and Makul [44] concluded that mixtures made with 20% RHA as partial replacement to OPC showed similar compressive strength compared to control mixtures up to 91 days due to pozzolanic reaction and denser internal structure [45]. Safiuddin et al. [46] concluded that inclusion of RHA enhanced compressive strength up to 56 days due to pozzolanic activity and micro-filling ability of RHA. At 28 days, compressive strength was between 42.7 and 94.1 MPa, whereas it ranged between 44.9 and 98.4 MPa for various mixes. Maximum compressive strength was observed for 30% RHA content with 0.35 w/b ratio. Table 5 shows

Table 5

Compressive strength of RHA-SCC mixes studied by several researchers.

Study	RHA (%)	w/b	Super plasticizer (%)	Compressive strength (MPa)				
				7 d	14 d	28 d	56 d	91 d
Chopra et al. [40]	0	0.41	1	29.0	–	36.7	39.6	–
	10			32.6	–	41.2	46.4	–
	15			36.2	–	48.8	53.7	–
	20			30.4	–	40.2	53.0	–
Pai et al. [35]	62.35	0.31	1.8	11.3	12.9	14.1	–	–
Rahman et al. [54]	0	0.38	1.8	32.8	–	48.5	–	–
	20	0.5	3.5	37.2	–	42.9	–	–
	30			35.1	–	40.9	–	–
	40			28.1	–	33.5	–	–
Sua-iam and Makul [32]	0	0.22	2	55.9	–	65.0	–	82.8
	10	0.31		48.4	–	54.8	–	72.6
	20	0.46		21.2	–	28.0	–	39.6
	40	0.75		13.8	–	19.1	–	26.4
	60	1.17		8.3	–	10.4	–	14.8
	80	1.80		2.8	–	4.1	–	5.7
	100	2.18		1.5	–	2.0	–	2.6
Memon et al. [45]	0	0.4	3.5	10.5	–	28.4	–	–
	5	0.38		25.2	–	38.0	–	–
	10	0.36		22.5	–	36.2	–	–
	0	0.4		6.8	–	18.3	–	–
	5	0.38	45	21.4	–	37.8	–	–
	10	0.36		36.8	–	41.4	–	–
	0	0.4		1.2	–	8.6	–	–
	5	0.38		11.9	–	22.2	–	–
	10	0.36		38.3	–	48.5	–	–
	0	0.39	2.2	36.5	37.6	37.8	–	44.7
	15	0.54		22.7	29.6	39.8	–	42.5

the compressive strength of RHA-SCC mixes reported by several researchers.

Chopra et al. [40] achieved compressive strength in the range between 36.7 and 41.2 MPa at 28 days and 39.6–46.4 MPa at 56 days for SCC mixes containing RHA (10–20%) with w/b ratio of 0.41. Maximum value of compressive strength was achieved with 15% RHA mix, whereas 20% RHA mix strength was greater than control SCC mix at 56 days of curing. The results were in concomitant with the studies reported by Kannan and Ganesan [42] and Chik et al. [47] where SCC mixes incorporating 15% RHA attained better 28-day strength than control SCC. Increased strength was due to highly reactive RHA particles that react with water and calcium hydroxide to produce additional calcium silicate hydrates (CSH) resulted in denser microstructure of concrete [40,44,46].

Atan and Awang [36] determined the compressive strength development in SCC incorporating raw RHA (RRHA), in binary, ternary and quaternary mixes with other types of mineral additives (LP, FA, and SF). All the mixes after 90 days of curing exhibited lower compressive strength compared to control. In binary mix (OPC + 15% RRHA), the strength development increases at high rate but less as compared to control mix due to reduced heat of hydration by replacing OPC with RRHA. It was observed that high silica content (92.99%) and specific area of RRHA resulted in comparable 90-day compressive strength in comparison with control mix. Inclusion of LP and FA in binary mix (OPC + RRHA + LP/FA) also produced comparable strength compared to control mix whereas mix containing SF showed reduced strength values. Similar observations were also made in quaternary mixes where mix containing OPC + RRHA + LP + FA produced lower strength values compared to other mixes and control mix. Juma et al. [48] found that incorporation of RHA (0–10%) and sugarcane bagasse ash (SCBA) alone and in blended with SCC significantly, increased compressive up to 28 days. The enhancement in strength was due to the smaller particle size and pozzolanic activity of RHA and SCBA that filled the micro-voids within the matrix.

Sua-iam and Makul [32] incorporated different proportions of RHA (0–100%) as a replacement of fine aggregate (river sand) and observed that by increasing the RHA content, strength decreased. At 28 days, SCC made with 100% RHA showed 3% of the compressive strength of the control mix whereas mix containing 10% RHA showed 84% of the strength. Similar observations were also made by Sua-iam and Makul [33] where increasing RHA content from (25–100%) the compressive strength decreased from 31.8, 54.5, 77.1 and 87.6%, respectively. Fig. 7 shows the SEM micrograph of internal structure of SCC control and SCC containing 25% RHA. The reduction in strength was due to coarse nature of RHA that decreased the packing density of matrix and made the SCC more porous (Fig. 7). Addition of supplementary minerals like limestone powder and coal fly ash along with RHA particles improved the compressive strength due to filler effect of mineral additives by improving pore structure and packing density [32,33].

Kannan and Ganesan [42,38] investigated SCC mixtures containing RHA (0, 5, 10, 15, 20, 25 and 30%) and SCC mixes incorporating RHA and MK in equal proportions (0, 10, 20, 30 and 40%) as partial replacement to OPC at 7, 28 and 90 days. The SCC mixture was designed to achieve targeted strength of 38.5 MPa (M30 grade) at 28 days with w/b ratio of 0.55 and 2% super plasticizer content. In binary blends, higher strength was observed in SCC containing 15% RHA at all ages. At 90 days, ternary blends (SCC + RHA + MK) exhibited slightly increased compressive strength than binary blends (SCC + RHA). Mixtures containing 15% RHA and 30% RHA + MK in equal proportions exhibit highest compressive strength values of 51.03 and 55.67 MPa, respectively, compared to control SCC (43.40 MPa) at 28 days. It was concluded that higher strength in RHA + MK mixes was due to high content of Al_2O_3 present in MK in comparison to RHA that accelerated the hardening process while presence of silica (SiO_2) in RHA reacted with CaO and supplemented the process of hardening in the alkaline environment [49]. On the contrary, Zhang and Malhotra [10] and Bhanumathi-

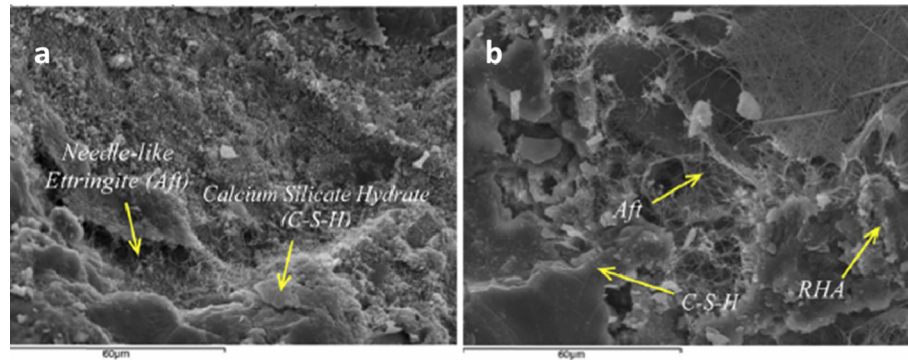


Fig. 7. SEM micrograph of internal structure at 28 day (a) SCC control, (b) SCC + 25%RHA [33].

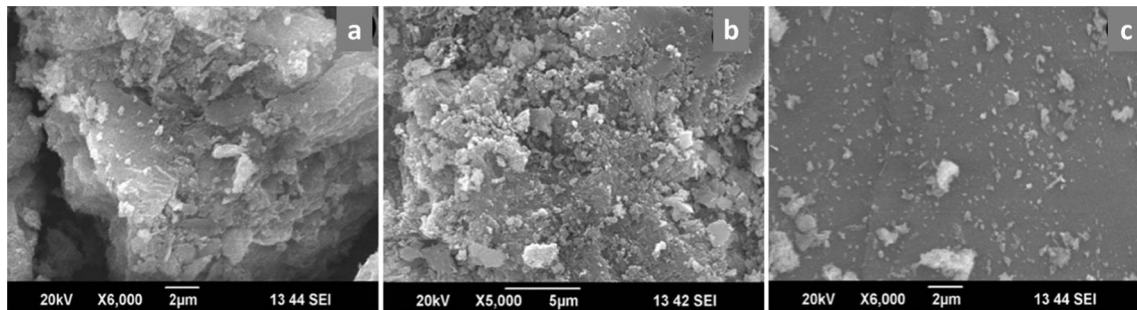


Fig. 8. SEM images of (a) control SCC; (b) 10% RHA-; (c) 30% RHA + MK- blended SCC [38].

das and Mehta [50] reported that up to 30% RHA can be used to achieve higher compressive strength in normal concrete.

Fig. 8 [38] and Table 6 shows analysis of scanning electron microscopic (SEM) and energy dispersive X-ray (EDAX) of the control SCC, SCC + 15%RHA and 15%RHA + 15%MK mixes carried out by Kannan and Ganesan [38]. From Fig. 8a, it can be concluded that the control SCC sample has porous and irregular microstructure as gypsum and ettringite are formed which resulted in increased pores and decreased durability and strength in comparison to blended mixes. Because of extra hydration in SCC mixes made with RHA and MK (Fig. 8b and c), porosity of SCC mixes got reduced providing better uniform structure in comparison with unblended SCC. From the corresponding EDAX spectrum of the SCC blends (Table 6), higher silica content was evaluated in 10% RHA (23.23%) and 30% RHA + MK (39.84%) compared to that of control SCC (11.62%). Good pozzolanic reaction, improved strength and reduced porosity incorporating RHA and MK in cement was due to higher silica content.

Incorporation of RHA increased the super plasticizer amount in SCC mixes due to high specific area and cellular structure of parti-

cles to maintain the flowability [51]. All the mixes containing grounded RHA (20, 30 and 40%) as partial replacement to cement showed reduced compressive strength values in comparison to control mix. The compressive strength ranged between 25.5 and 27 MPa, which was higher than 20 MPa (design at the age of 28 days). Therefore, it was suggested that the use of RHA was effective in producing SCC with 20–30% of RHBA replacement. Pai et al. [35] evaluated the compressive strength of SCC mixes incorporating RHA and reported lower strength values all ages (7, 14 and 28 days) due to addition of high amount of RHA (62.35%; 331.3 kg/m³) compared to control mix.

Memon et al. [31,45] studied the compressive strength of SCC incorporating rice husk ash (5 and 10%) with three w/b ratios of 0.4, 0.38 and 0.36; and three percentages by weight of super plasticizer 3.5, 4 and 4.5. At 7 and 28 days, with 3.5% of super plasticizer content control mix attained maximum compressive strength of 10.5 and 28.4 MPa, but for control SCC and SCC with 5% RHA, by increasing the dosage of super plasticizer strength starts decreasing. Strength of SCC mixes made with RHA (10%) increased with increase in superplasticizer content. At similar super plasticizer dosage, RHA mixes showed higher compressive strength compared to control mixes. This increase was basically governed by reduced w/b ratio, improved microstructure, dense matrix, pore and grain size refinement. Sadrmomtazi and Barzegar [52] studied the effect of nanosilica (NS) on the compressive strength of SCC with RHA (20%) up to the age of 60 days, and observed that with addition of NS (7%), strength increased by 86, 64 and 58% after 7, 28 and 60 days as compared to control SCC (19.1, 30 and 35.4 MPa) whereas an increase of 51, 34 and 15% in strength was exhibited by NS-RHA-SCC mix when compared with RHA-SCC (19.9, 34.8 and 46.5 MPa) at similar curing periods. Use of nano-silica in SCC made cement paste thicker, resulting denser matrix of the concrete. Le and Ludwig [41] concluded in the study that with slightly finer particles (RHA 5.7 µm) than the cement

Table 6
EDAX analysis of SCC, 10%RHA, and 30% RHA + MK- blended SCC [38]

Element (Mass %)	Control SCC	10% RHA	30% RHA + MK
C K	4.9	1.68	–
O K	8.63	30.56	28.65
Na K	NA	NA	1.2
Mg K	3.83	9.59	NA
Al K	1.77	6.47	15.36
Si K	11.62	23.23	39.84
Ca K	61.96	1.14	NA
Fe K	3.02	2.56	NA
Cu K	4.27	24.77	NA
K K	NA	NA	14.95

particles (7.07 μm), RHA can yield a higher packing density in the granular mixture. Moreover, the increase in compressive strength of SCHPC (self-compacting high performance concrete) incorporating RHA at more matured conditions is also due to the internal water curing effect of RHA. RHA with porous structure may absorb free water during mixing. This amount of water is released into the cement paste surrounding the RHA particles, when the relative humidity in the paste diminishes during maturation facilitating a prolonged hydration. However, the incorporation of high content of RHA induced adverse effects on compressive strength at early ages (3 and 7 days). At higher replacement levels by RHA, the concrete will contain a significantly reduced cement content which accounts for the lower strength at 3 days. RHA behaves as hydrophilic material leads to lack of available water for cement hydration reaction. Moreover, RHA particles are themselves weakest points in the hardened matrix due to their pore structure. RHA was effective in improving compressive strength of SCHPC, particularly at larger percentage replacement and at late ages. Blended mixes of RHA and FA (Fly Ash) improved compressive strength and helps in achieving better SCC. Compressive strength of SCHPC incorporating 20 wt% FA and 20 wt% RHA reached about 130 MPa after 56 days. Ranjbar et al. [24] studied compressive strength, incorporating rice husk ash (RHA) and nano-TiO₂ (NT) in cement mortars. A total of 13 different mixtures with different amounts of RHA, NT and SP were prepared with total binder contents of 700 kg/m³, for all mix proportions. The percentage of RHA was varied between 0 and 15% by weight of the total binder. The percentage of nanoparticles was 0, 1, 3 and 5% of the binder. The amount of SP varied between 0.6 and 1.2% by weight of the binder. The water to binder ratio (w/b) was kept constant at 0.4 for all the mixtures. The mixture proportions of the ingredients were calculated by the volumetric method as specific gravity of RHA and NT is lower than that of cement, use of RHA and NT in place of cement by weight percentage reduced the amount of sand to maintain the same volume. Compressive strength results illustrated a substantial improvement in samples containing NT and also a slight increase in mortar performance was observed by using up to 10 wt% of RHA as a replacement of cement. Alex et al. [53] studied compressive strength and found that 20% RHA replacement with cement is optimum. The increase in specific surface area of RHA contributes to the increased consumption of Ca(OH)₂ in concrete i.e., the RHA with high specific surface area shows high reaction rate between calcium hydroxide and the silica, thereby exhibiting exceptional pozzolanicity and helps to increase the mechanical strength of the concrete. On studying the effect of the RHA dosage in concrete strength development found that 10% cement replacement showed a remarkable percentage of strength gain (7.8%) as compared to normal concrete. The strength increase is due to the higher content of calcium silicate hydrate (CSH) in the RHA blended concrete specimens, due to the reaction of the calcium hydroxide produced from cement hydration with the active silica of the RHA. The CSH is the main carrier of strength in hardened cement. However, 20% replaced concrete also performed better than the normal concrete, thereby contenting to be the optimal replacement level.

It can be concluded that higher content of RHA increases compressive strength because of its micro-filling and pozzolanic

effects, which improve the microstructure of concrete in bulk paste, and 15% RHA is the optimum content.

3.2. Flexural strength

Table 7 shows the flexural strength of RHA-SCC mixes reported by several researchers. Ahmadi et al. [43] investigated the flexural strength of SCC mix containing RHA (10 and 20%) at different ages from 7 to 180 days in comparison to normal concrete at two w/b ratios (0.40 and 0.35). It was observed that the SCC mixes exhibited flexural strength about 12–20% more than normal concrete. Increasing the RHA content increased the flexural strength after 60 days of curing due to increase in pozzolanic reaction of RHA in the matrix and highest flexural strength in all cases was observed in mixes containing 20% RHA. Atan & Awang [36] reported increased flexural strength in 15% RRHA + OPC mix (6.5 MPa) compared to control mix (5.7 MPa) whereas addition of mineral additives (FA and SF) in equal mass to binary blends produced substantially lower flexural strengths but comparable to control mix. On the contrary, inclusion of LP increased the flexural strength in ternary (OPC + RRHA + LP) and quaternary (OPC + RRHA + LP + SF) blends due to pore-filling effect of LP which densifies the concrete microstructure. Pai et al. [35] also concluded that addition of large amount of RHA (62.35%) in SCC negatively affect the strength of concrete. Incorporation of 3% nanosilica (NS) enhanced the flexural strength of RHA-SCC by 50.7, 33.6 and 15% compared to RHA alone mix at 7, 28 and 60 days, respectively [52].

3.3. Splitting tensile strength

Rahman et al. [54] studied the splitting tensile strength of SCC containing RHA (0, 20, 30 and 40%) as partial cement replacement. The results showed that splitting tensile strength decreases with increase in content of RHA. Incorporation of 20% RHA produced strength values similar to that of control mix. Chopra et al. [40] achieved tensile strengths in the range 2–2.8, 2.5–3.7 and 2.8–4 MPa at 7, 28 and 56 days, respectively, in SCC containing 10–20% RHA; and observed that up to 15% replacement splitting tensile strength increased but the strength in 20% RHA mix were acceptable as strength was still higher than control mix (Table 8).

Khadiy et al. [55] aimed at producing and comparing SCC incorporating rice husk ash (RHA) and shell lime powder (SL), both locally available mineral admixtures as an additional cementing material. At 28 days, higher strength for RHA mix was observed in comparison to SL. Splitting tensile strength of SL-SCC was 23.98 and 5.2% higher than RHA-SCC for 7 and 14 days of curing, whereas for 28 days of curing, the strength of RHA-SCC was 0.8% higher than that of SL-SCC. Better strength in RHA in comparison with SL is possibly because silica content in RHA reacts better with cement than calcite present in SL.

Based on literature it has been found that 15–20% RHA as partial replacement of cement is acceptable as RHA does not significantly changes the compressive strength.

Table 7
Flexural strength of RHA-SCC mixes studied by several researchers.

Study	RHA replacement (%)	w/b	Super plasticizer content (%)	Flexural strength (MPa)				
				7 d	14 d	28 d	56 d	91 d
Pai et al. [35]	62.35	0.31	1.8	2.3	3.3	3.8	–	–
Atan and Awang [36]	0	0.39	2.2	4.5	4.7	5.7	–	5.7
	15	0.54		3.2	3.5	4.0	–	6.5

Table 8

Splitting tensile strength of RHA-SCC mixes studied by several researchers.

Study	RHA replacement (%)	w/b	Super plasticizer content (%)	Splitting tensile strength (MPa)			
				7 d	14 d	28 d	56 d
Chopra et al. [40]	0	0.41	1	2	–	2.5	2.8
	10			2.4	–	3.6	3.8
	15			2.8	–	3.7	4.0
	20			2.3	–	3.0	3.3
Rahman et al. [54]	0	0.38	1.8	–	–	5.1	–
	20	0.5	3.5	–	–	5.1	–
	30			–	–	4.3	–
	40			–	–	2.8	–
Pai et al. [35]	62.35	0.31	1.8	0.7	0.9	1.1	–

3.4. Modulus of elasticity

Ahmadi et al. [43] observed modulus of elasticity of SSC containing RHA increased with age similar to compressive and tensile strength. Normal concrete modulus of elasticity was 9–17% higher than SCC mixes. Further, modulus of elasticity of SCC mixes reduces with increase in percentage of RHA.

4. Effect of RHA on durability properties of SCC

4.1. Ultrasonic pulse velocity (UPV)

An UPV measurement is directly proportional to that of compressive strength, with increase in compressive strength UPV also increases. Sua-iam and Makul [32] found that UPV in SCC containing RHA (0, 10, 20, 40, 60, 80 and 100%) as replacement to fine aggregate varied in the range 0.7–4.2 km/s and 1–5 km/s at 28 and 91 days, respectively, compared to 4.4 and 5.2 km/s of control mix (Table 9). The higher velocity, indicated better quality of SCC, was achieved in control mix while 100% RHA mix achieved lowest pulse velocity. It was observed that with increasing RHA values UPV decreases. Sua-iam and Makul [33] reported 7.8, 25.5, 42 and 62.8% decrease in UPV in SCC mixtures with 25, 50, 75 and 100% RHA. Addition of mineral additive such as limestone powder and fly ash enhances the UPV in SCC due to denser microstructure.

Incorporation of RHA (0–30%) as partial replacement to cement with varying w/b ratio of 0.30, 0.35, 0.40 and 0.50 showed the UPV in the range 4.730–5.097 km/s and indicated the excellent condition of SCC mixtures [46]. Highest level of UPV was achieved for concretes with w/b ratio of 0.30, whereas with w/b ratio of 0.50 lowest level of UPV was obtained. Due to microfilling ability and pozzolanic effect of RHA, the pore refinement and porosity reduction lead to increase pulse velocity of SCC-RHA mixes. It was also observed that SCC-RHA mixes with air content (2%) showed higher UPV than mixes containing 6% air content. The entrained air voids and pores in the interface delayed the propagation of the ultrasonic pulse, thus reducing the UPV of concrete. Ranjbar et al. [24] in his study concluded the UPV for RHA and nano-TiO₂ (NT) blended

mortars experienced a sharp increase compared to the small proportion obtained by the control sample. Further, incorporating RHA without NT increased the UPV by 8% in comparison to the control sample. Also, incorporating NT at 5% content can rise the UPV noticeably. The result illustrated that 15% and 5% is the optimal proportion of the RHA and NT replacement in the mortar, respectively. The UPV test revealed that the permeability was reduced with an increase in the contents of RHA and NT with respect to the control specimen. It was concluded that a combination of 15% RHA and 5% NT in mortar led to a positive contribution to durability properties at 90 days.

It concluded that by incorporating RHA in SCC UPV of concrete increases due to its pore refinement and denser matrix making concrete more durable.

4.2. Rapid chloride permeability test

Ramasamy [56] concluded that inclusion of RHA as partial replacement of cement results in increased values of chloride ion permeability values (charge passed) but were in the “very low category” (100–1000 coulombs). Similar behaviour was illustrated by Chopra et al. [40] in RHA replaced SCC where decrease in charge passed was observed in mixes containing up to 15% RHA. In 20% RHA, the charge passed increased but remained lower than control mix. Zhang et al. [57], and Zhang and Malhotra [10] reported excellent resistance by 10% RHA concrete and the value of charge passed was lying in very low category both at 28 and 91 days.

Use of RHA in concrete refines the pore structure resulting in reduced chloride ion permeability [56,40]. Kannan and Ganesan [42] studied the chloride ion permeability in SCC mixes with MK, RHA and RHA + MK according to ASTM C 1202 [58] after 28 days of curing. The minimum chloride ion permeability was observed in mixes blended with 15% RHA (306.22 coulomb), 30% MK (28.23 coulomb) and 40% RHA + MK (25.43 coulomb) compared to control SCC specimens (1486.28 coulomb). The total charge passed values of 30% MK (28.23 coulomb) and 40% RHA + MK (25.43 coulomb) SCC specimens were categorized under “very low” chloride ion permeability and this was due to fineness of MK particles, and pozzolanic reaction of MK with Portlandite (Ca(OH)₂) resulted in reduced pore network and exhibited a much better resistance to chloride ion penetration (Table 10). Ranjbar et al. [24] concluded that by using both RHA and TiO₂ nano particles have positive impact on the chloride permeability results of the mixtures. It is also obvious that the charge passed in the binary mixtures containing RHA and nano-TiO₂ (NT) were less than that in the control mixture. The least values of charge passed were obtained by mixtures with 15% RHA. The decrease in the pore interconnectivity of mortars and pore water solution has led to reduce the permeability of RHA mortar.

Permeability of SCC decreases considerably with the use of RHA due to improved particle packing behaviour makes SCC denser.

Table 9

Ultrasonic pulse velocity of SCC mixes containing RHA [32]

Mix	Ultrasonic pulse velocity (km/s)			
	1 d	7 d	28 d	91 d
Control (0%)	2.5	3.6	4.4	5.2
10% RHA	2.1	3.4	4.2	5.0
20% RHA	2.0	2.8	3.5	4.2
40% RHA	0.8	1.4	2.2	3.0
60% RHA	0.7	1.1	1.8	2.6
80% RHA	0.5	0.7	0.9	1.3
100% RHA	0.4	0.6	0.7	1.0

Table 10
RCPT of SCC mixes containing RHA.

Mix	RCPT (coulombs) at the age of 28 d	
	Chopra et al. [40]	Kannan and Ganesan [42]
Control (0%)	2830	1486
5% RHA	1970	439
10% RHA	980	389
15% RHA	1173	306
20% RHA	–	877
25% RHA	–	905
30% RHA	–	1089

This is because RHA is silica rich, reacts with calcium hydroxide forming additional C–S–H gel which improves the microstructure by filling the pores and leads to considerable improvement in durability than control mix.

4.3. Water absorption and porosity

Kosmatka et al. [59] observed that good quality concrete should have water absorption less than 5% and this can be achieved by increasing density and reducing pore size so that there is poor pore connectivity and decreased porosity of concrete. Safiuddin et al. [46] investigated that water absorption in SCC-RHA mixtures at w/b ratio of 0.30–0.50 varied in the range 2.89–5.97%, which was relatively low. Water absorption decreased at w/b ratio of 0.30 and was about 25% as compared with w/b ratio of 0.50. Inclusion of RHA reduced the water absorption and least value of water absorption observed was for 30% RHA concrete at 0.35 w/b ratio which was 35% less than 0% RHA concrete at similar w/b ratio. Similarly, Safiuddin et al. [46] observed the total porosity in the range 6.77–13.71%. The total porosity decreased with increase in RHA content and reduction in 5–35% of total porosity was observed for various RHA contents.

Table 11 shows the water absorption and porosity of SCC mixes containing RHA studied by several researchers. Rukzon and Chindaparasirt [51] and Chopra et al. [40] reported that with curing period the porosity of RHA-SCC mixes reduced due to pozzolanic reaction. Mixtures up to 20% RHA exhibited lower porosity values than control SCC mixes at 28 days. These results can be compared with the findings of Ramasamy [56] where the porosity values decreased (4.7, 4.5, 4.2, 3.9 and 3.45%) with increase in RHA content (0, 5, 10, 15 and 20%) for M30 concrete mixes without the use of super plasticizers. Also, addition of super plasticizers the porosity values varied in the range 3.80–5.20%.

Kannan and Ganesan [42] studied the effect of RHA and MK (5–30%) and RHA + MK (10–40%) as partial replacement of cement on the water absorption (WA) of SCC at 28, and observed that with increasing MK and RHA + MK content, water absorption decreases except up to 15% RHA content compared to control SCC (4.54%

Table 11
Water absorption and porosity of SCC mixes containing RHA at the age of 28 days.

Mix	Water absorption (%)		Porosity (%)
	Rahman et al. [54]	Kannan and Ganesan [42]	Chopra et al. [40]
Control (0%)	6.2	4.5	12.4
5% RHA	–	4.5	10.8
10% RHA	–	4.1	10.0
15% RHA	–	3.9	11.1
20% RHA	7.7	3.9	–
25% RHA	–	4.5	–
30% RHA	8.9	4.9	–
40% RHA	10.5	–	–

WA). Higher water absorption capacity of SCC mixes made with more than 15% RHA, is due to higher surface area and lower fineness of RHA, resulting in increased water quantity, reduced workability which subsequently create voids. Sadrmomtazi and Barzegar [52] observed that incorporation of nanosilica (NS) reduced the water absorption in 20% RHA-SCC mix from 5.33 to 4.31% due to formation of nucleation sites for hydration process by NS, increased CSH, reduced porosity and by act as filler material that block the capillary pores and water channels in cement paste [60,61]. Ranjbar et al. [24] concluded the proportion of water absorption reduced considerably by adding nano-TiO₂ (NT) and RHA. The partial replacement of Portland cement by 15 wt% RHA and 5 wt% NT led to an almost 30% reduction in water absorption. The result could be related to the fact that since the pozzolanic reaction has consumed Ca(OH)₂ creating more C–S–H this has led to a denser microstructure and consequently less water absorption. The reduction in water absorption with the increase of NT content in the mixtures is resulted from enhancing the pore structure. Another important reason is the role of NT as filler densifying the microstructure and the ITZ, causing a reduced porosity.

Incorporating RHA up to 15% in SCC significantly reduces water absorption and is primarily credited to improved microstructure.

4.4. Sorptivity

Kannan and Ganesan [42] investigated the sorptivity of SCC mixes with MK, RHA and RHA + MK. Increasing RHA content up to 15%, the sorptivity decreases from 3.56 (control SCC) to $3.31 \times 10^{-6} \text{ m/s}^{1/2}$ whereas with MK and RHA + MK the minimum sorptivity (2.29 and $2.64 \times 10^{-6} \text{ m/s}^{1/2}$, respectively) was observed for 20 and 30% replacement levels. In MK blended SCC, the minimum sorptivity was due to finer size of MK than OPC and RHA and increased formation of calcium silicate hydrate (CSH) gel leads to a reduction in pores size and ultimately the sorptivity. The increased sorptivity values in 20% ($4.06 \times 10^{-6} \text{ m/s}^{1/2}$), 25% ($6.41 \times 10^{-6} \text{ m/s}^{1/2}$) and 30% RHA ($9.2 \times 10^{-6} \text{ m/s}^{1/2}$) was due to pores developed in concrete due to reduced workability. At 28 days, it was also observed that SCC specimens blended with 30% MK and 40% RHA + MK showed 18.82% and 19.10% reduction in sorptivity respectively, compared to control SCC ($3.56 \times 10^{-6} \text{ m/s}^{1/2}$).

4.5. Electrical resistivity

At 28 and 56 days, electrical resistivity of concrete varied between 4.1 and 121.2 kΩ cm [46]. Mixtures that exhibits electrical resistivity between 5 and 10 kΩ cm show moderate to low corrosion rate, however corrosion resistance concrete is considered good when electrical resistivity is above 10 kΩ cm [62]. All RHA-SCC mixtures exhibited a true electrical resistivity higher than 10 kΩ cm compared to non-RHA-SCC mixtures (4.1–8.9 kΩ cm). High w/b ratio of 0.50 showed lowest level of true electrical resistivity whereas w/b ratio of 0.30 showed highest level of true electrical resistivity due to the pore refinement, improved microstructure with a reduced porosity. Ranjbar et al. [24] concluded significant improvements in electrical resistance of mortars were achieved with the increase of TiO₂ nanoparticles and RHA in the mixture. Low or medium probability of corrosion was observed in other samples having resistivity values ranging between 10 and 20 kΩ cm. The optimum content of RHA and NT in the cement mortar were 15% and 5%, respectively.

4.6. Acid resistance

The acid resistance of SCC mixes blended with MK, RHA and RHA + MK was assessed by Kannan and Ganesan [42] at 28 days.

The weight loss of blended SCC mixes was measured for 12 weeks with 1 week interval after immersed in 5% sulfuric acid (H_2SO_4) and hydrochloric acid (HCl) solution. It was observed that RHA and RHA + MK – SCC blends exhibited better resistance against sulfuric acid attack than MK–SCC blends. Weight loss observed was lowest in 25% RHA, 5% MK and 40% RHA + MK replacement levels. The maximum weight loss and deterioration was observed in MK blended SCC due to presence of high alumina content in MK. Calcium sulfoaluminate (ettringite) is formed when Al_2O_3 reacts with silica SiO_2 , resulting in expansion in concrete, and disrupting the set cement paste. However in RHA and RHA + MK mixes, there was lesser formation of ettringites because there was less alumina content in comparison to MK. Similar trend in results was achieved in 25% RHA, 5% MK and 40% RHA + MK replacement levels when immersed in hydrochloric acid (HCl) solution. MK blended SCC was less resistance to hydrochloric acid compared to RHA and RHA + MK blended SCC. Weight loss for all mixtures in the hydrochloric acid solution was lower than in the sulfuric acid solution.

4.7. Alkali silica reaction

Le et al. [63] investigated the resistance of mortars formulated from SCC containing RHA to alkali silica reaction. The effect of RHA (particle size 5.7, 7.37 and 15.6 μm) at 20% on ASR expansion of mortar bars ($40 \times 40 \times 160$ mm) with reactive greywacke aggregates was evaluated after 28 day immersion in 1 M NaOH solution at 80 °C. After 14 days, specimens showed expansion lower than 0.10% whereas after 28 days, control sample and RHA (15.6 μm) sample exhibited increase in expansion of 0.27 and 0.46%, respectively. Several visible cracks were also visible on the surface of RHA samples with particle size 7.7 and 15.6 μm , and the intensity of the cracks increased with increase in particle size. Mortar samples with RHA particle size 7.7 μm showed higher expansion after 56 days of immersion in 1 M NaOH solution at 80 °C and at later age samples with RHA particle size 5.7 μm followed the same trend, this may be because of its porous structure.

5. Summary and conclusions

Following conclusions can be drawn from the published literature.

- High silica content in RHA makes it suitable pozzolanic construction material for long term strength development in SCC.
- Compressive strength, splitting tensile strength and flexural strength of SCC increases with increase in RHA content up to 15% because of its micro-filling and pozzolanic effects, which improve the microstructure and pore structure of concrete.
- Increasing RHA content in SCC reduces the workability. To achieve more workable and higher strength superplasticizer should therefore be added to the concrete mixtures incorporating RHA.
- Water absorption and Porosity values decreases with the increase in RHA content up to 30% (cement replacement) because small RHA particles improved the particle packing density of blended cement leading to a reduced volume of larger pores.
- XRD analysis of RHA depicts that C–S–H formation increases with increase of RHA content in SCC, which is the main reason for increase in strength as compared to other mixes. SEM shows that the RHA sample is multi-dispersed with micro porous surface and irregular shaped particles.
- High silica content in RHA makes it suitable pozzolanic constructional material for long term strength development in SCC.
- Incorporation of RHA in SCC and blending with other mineral additives (fly ash, silica fume, limestone, metakaolin, pulverized fuel ash, etc.) may prove beneficial in improving the strength and durability properties.
- Replacement of RHA in both fine aggregates and cement replacement reduces the cost of making concrete.

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